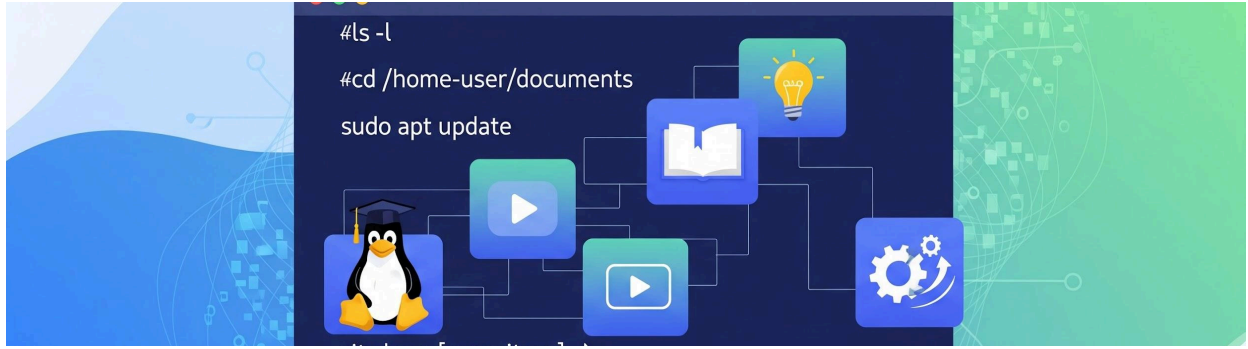




LINUX OS

Introduction:

- Linux is a free and open-source operating system with high security.
- Linux is a multi-user operating system.

OPERATING SYSTEM

- An Operating System (OS) is software that acts as an interface between computer hardware components and the user.
- Every computer system must have at least one operating system to run other programs.
- Applications like browsers, MS Office, Notepad, games, etc., need an environment to run and perform tasks.
- The OS helps users communicate with the computer without knowing the computer's language.

KERNEL

- The kernel manages the hardware components such as CPU, memory, and peripheral devices.
- The kernel is the lowest level of the operating system.

DAEMONS

- Daemons manage background services such as printing, sound, scheduling, etc.
- These services start during system boot or after user login.

SHELL

- The shell is an environment where users can run commands, programs, and shell scripts.
- It takes input from the user, executes it, and displays the output.
- A shell acts as an interface between the user and the operating system.

Shells can be:

Ø Graphical (GUI-based)

Ø Command-line based (CLI)

COMMAND

- A command is an instruction or request given to the operating system by a user.
- It tells the computer to perform a specific task.

TERMINAL

- A terminal is a text-based interface that allows users to interact with the operating system by typing commands and receiving output.
- Terminals are commonly used in Linux, Unix, macOS, and Windows systems.

LINUX OS DISTRIBUTIONS

- Linux distributions are modified versions of Linux released with different names based on user requirements.

Examples:

- i. RedHat
- ii. Ubuntu
- iii. Debian
- iv. CentOS
- v. Fedora
- vi. OpenSUSE
- vii. Kali Linux
- viii. Amazon Linux

ix. Rocky Linux

LINUX HISTORY

- In 1991, Linus Torvalds, a student at the University of Helsinki, Finland, started developing Linux.
- The Linux kernel is written in C language.
- Initially, the name was planned as Freax, but later it became Linux.
- In 1992, Linux was released under the GNU General Public License.

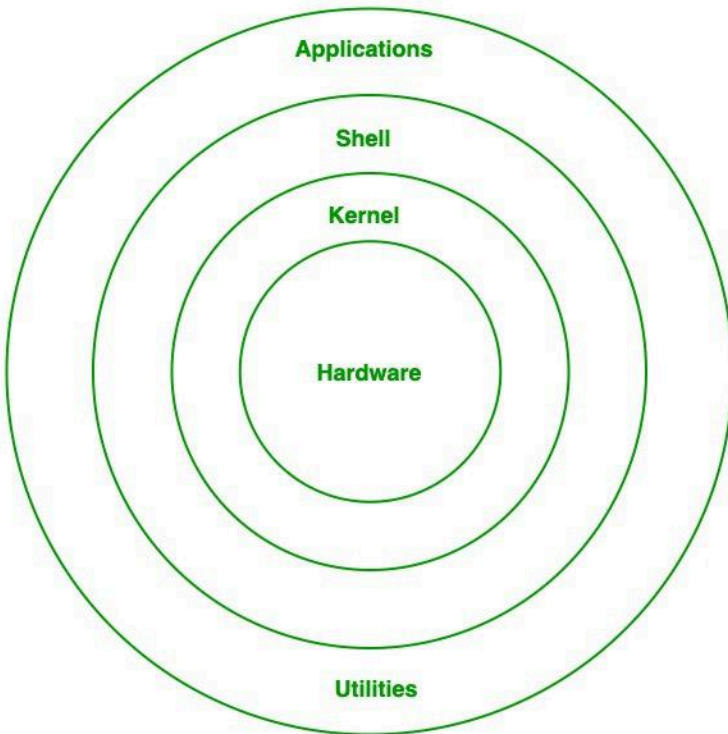
OPEN SOURCE

- Linux is distributed under an open-source license.
- Open source provides:
 - Freedom to use the software
 - Freedom to study and modify the code
 - Freedom to redistribute copies
 - Freedom to distribute modified versions

ARCHITECTURE

The Linux architecture consists of:

- Kernel
- Shell
- Commands
- Terminal



Introduction to Essential Linux Commands

Linux is a powerful and popular operating system, especially for servers, developers, and advanced users. To effectively use Linux, you interact with it through the **Command Line Interface (CLI)**, also known as the terminal or shell.

This document provides a detailed, yet simple, guide to essential Linux commands.

Basic Navigation and File Management

These commands help you move around the filesystem and handle files and directories (folders).

Command	Simple Language Explanation	Example
<code>pwd</code>	P rint W orking D irectory. Shows you the full path of the directory you are currently in.	<code>pwd</code>
<code>ls</code>	L ist. Shows you the files and folders inside your current directory.	<code>ls</code>
<code>ls -l</code>	A "long list" that shows detailed information (permissions, size, owner, date).	<code>ls -l</code>
<code>cd</code>	C hange D irectory. Used to move from one directory to another.	<code>cd Documents</code> (moves into the Documents folder)
<code>cd ..</code>	Moves you up one level (to the parent directory).	<code>cd ..</code>
<code>cd ~</code>	Moves you directly to your user's Home directory (the starting point).	<code>cd ~</code>
<code>mkdir</code>	M ake D irectory. Creates a new folder.	<code>mkdir new_project</code>
<code>touch</code>	Creates a brand-new, empty file.	<code>touch notes.txt</code>

File Operations

Once you have files, you need commands to view, copy, move, and delete them.

Command	Simple Language Explanation	Example
<code>cat</code>	C atenate. Shows the entire content of a file directly in the terminal.	<code>cat notes.txt</code>

Command	Simple Language Explanation	Example
<code>less</code>	Views file content page-by-page, good for large files. Press <code>q</code> to quit.	<code>less very_large_log.txt</code>
<code>cp</code>	Copy. Copies a file from one location to another.	<code>cp file1.txt /home/placeholder-person/backup/</code>
<code>mv</code>	Move. Moves a file (like cutting and pasting). Also used to rename files.	<code>mv oldname.txt newname.txt</code>
<code>rm</code>	Remove. Permanently deletes a file (be careful, there is no Recycle Bin).	<code>rm temp.log</code>
<code>rm -r</code>	Removes an empty or non-empty directory. The <code>-r</code> stands for recursive.	<code>rm -r old_folder</code>

Getting Help

The Linux command line has built-in help features for almost every command.

Command	Simple Language Explanation	Example
<code>man</code>	Manual. Shows the comprehensive reference manual page for a command.	<code>man ls</code>
<code>--help</code>	A simple option available on most commands that provides a quick list of options.	<code>ls --help</code>

Viewing System Information

These commands provide insights into how your system is running.

Command	Simple Language Explanation	Example
<code>date</code>	Displays the current system date and time.	<code>date</code>
<code>whoami</code>	Tells you the name of the user you are currently logged in as.	<code>whoami</code>
<code>uname -a</code>	Displays detailed information about your operating system kernel.	<code>uname -a</code>
<code>df -h</code>	Disk Free. Shows the amount of available disk space, in "human-readable" format.	<code>df -h</code>
<code>top</code>	Shows a real-time list of running processes, like the Windows Task Manager. Press <code>q</code> to quit.	<code>top</code>

Search and Location

Commands to find files or text within files.

Command	Simple Language Explanation	Example
<code>grep</code>	Global Regular Expression Print. Searches for specific text inside files.	<code>grep "error" server.log</code> (finds the word "error" in the log file)
<code>find</code>	Searches for files and directories based on various criteria (name, size, date).	<code>find . -name "*.txt"</code> (finds all files ending in <code>.txt</code> in the current directory)

Permissions

Linux uses permissions to control who can read, write, or execute a file.

Command	Simple Language Explanation	Example
<code>chmod</code>	Change Mode. Changes the permissions of a file or folder.	<code>chmod +x script.sh</code> (makes a file executable)
<code>chown</code>	Change Owner. Changes the owner of a file or folder.	<code>chown :users important_file.conf</code>

For further details on any specific command, please refer to the official documentation or use the `man` command. A list of advanced commands is available in the document titled [File](#).